Cambridge O Level Computer Science

Chapter 4: Software

Lecture notes with Check-Yourself questions and a full answer key.

In this chapter you will learn about:

- Types of software and interrupts: system software vs application software (operating systems, utility programs, device drivers), the role and functions of an operating system, booting and firmware, and the role and operation of interrupts.
- Programming languages, translators and IDEs: high-level vs low-level languages, assembly language, compilers, interpreters and assemblers, and the features of integrated development environments.

Without software, computer hardware would be useless. This chapter first covers how software is used, from essential system software to everyday applications, and then how programs are translated so the computer can carry out their instructions.

4.1 Types of software and interrupts

4.1.1 System software and application software

There are two types of software, sitting in a hierarchy with hardware at the bottom, system software above it, and application software on top:

- System software: programs that control and manage the computer hardware. It provides a platform for other software to run, lets hardware and software work together without problems, provides a human-computer interface (HCI), and controls the allocation and use of hardware resources. Examples: operating systems, utility programs, device drivers, compilers and linkers.
- Application software: programs that let the user perform specific tasks using the computer's resources. It can be a single program (for example, NotePad) or a suite (for example, Microsoft Office), and the user runs it whenever needed.

Typical application software: word processors (create, edit, save and format text; copy and paste; spell checker and thesaurus; import images; translate languages); spreadsheets (organise numbers in a grid of cells such as B6; formulas, graphs, modelling and what-if calculations); databases (tables of records and fields;

sakib sir

queries, reports, adding, deleting and modifying data); control and measuring software (interfaces with sensors to measure quantities and control processes, for example opening a valve to change pH); photo editing (brightness, contrast, red-eye removal, retouching); video editing (rearranging clips, titles, colour correction, transitions); graphics manipulation (bitmap editors change pixels; vector editors manipulate lines, curves and text); apps (mobile and tablet software from an app store: streaming, GPS, camera).

Typical system software: compilers (translate high-level language to machine code); linkers (combine compiler-produced modules into one runnable program); device drivers (let hardware communicate with the operating system); operating systems; utilities.

Check yourself

1. 1. State two differences between system software and application software.
2. 2. Classify each of these as system software or application software: (a) word processor, (b) device driver, (c) anti-virus program, (d) video editing software, (e) operating system.
3. 3. Give two functions of spreadsheet software and two functions of database software.
4. 4. What is the difference between how a bitmap graphics editor and a vector graphics editor change an image?

Utility programs

Utility programs (utilities) are system software that manage, maintain and control computer resources. They are often started by the user, though some, notably virus checkers, run in the background without user input.

Virus checkers (anti-virus software)

Any computer, including phones and tablets, can suffer a virus attack, so anti-virus software should run in the background. Common features of all anti-virus software:

- Checks software and files before they are run or loaded.
- Compares possible viruses against a database of known viruses.
- Carries out heuristic checking: looking for behaviour that suggests a virus, useful against viruses not yet in the database.

sakib sir

- Puts infected files into quarantine, where they can be automatically deleted or the user decides (a file flagged by mistake is a false positive).
- Must be kept up to date, since new viruses appear constantly.
- Full system scans should run regularly (for example, weekly), since some viruses lie dormant and only a full scan finds them.

Prevention also helps: be careful downloading from the internet, do not open files or links in emails from unknown senders, and avoid non-original software. But virus checkers remain the best defence.

Defragmentation software

As a hard disk fills, file blocks scatter across different sectors and tracks (files are deleted, partly deleted or extended), so the read-write head must move many times to retrieve one file and access slows down. A disk defragmenter rearranges the blocks so each file occupies contiguous (next-to-each-other) sectors, greatly reducing head movements and speeding up access. (This is not an issue for SSDs, which access data differently.)

Worked example: A disk track has 8 sectors (0 to 7). File A occupies sectors 0, 1 and 4; file B occupies sectors 2 and 3. Show the layout after defragmentation.

Before: sectors 0:A, 1:A, 2:B, 3:B, 4:A, 5:free, 6:free, 7:free. File A is split.

The defragmenter moves file A's scattered block into contiguous sectors: 0:A, 1:A, 2:A.

File B follows contiguously: 3:B, 4:B.

After: 0:A, 1:A, 2:A, 3:B, 4:B, 5:free, 6:free, 7:free. Each file is now contiguous, so fewer head movements are needed.

Back-up software

As well as manual back-ups (for example, to a memory stick), the operating system's back-up utility lets you schedule back-ups and only backs up files that have changed. For total security keep three versions of a file:

5. The current working version on the internal HDD or SSD.
6. A locally backed-up copy (for example, on a portable SSD).
7. A remote back-up copy well away from the computer (for example, cloud storage).

sakib sir

Windows File History takes snapshots of files onto an external HDD at regular intervals (by default every hour, kept forever unless changed), building a library of past versions to choose from, plus restore points that roll the whole computer back to an earlier state. Mac OS Time Machine erases a selected external drive and replaces it with the back-up: hourly back-ups, daily back-ups for the past month, and weekly back-ups for earlier months (oldest back-ups are deleted when the drive fills).

Security software and screensavers

Security software is an over-arching utility that: manages access control and user accounts (user IDs and passwords); links to virus and spyware checkers; protects network interfaces with firewalls; uses encryption and decryption so intercepted data is meaningless; and oversees software updates (checking they come from a legitimate source).

Screensavers show moving or still images after a period of inactivity. They were invented to stop phosphor burn on old CRT monitors; modern LCD and OLED screens do not need this, so screensavers are now mostly customisation. But they also serve security: after, say, five minutes idle, the screensaver loads and the computer locks, logging the user out. Idle time is also used for background tasks such as virus scans or distributed computing apps.

Device drivers

A device driver communicates with the operating system and translates data into a format a hardware peripheral understands. Without the right driver, a device cannot work and a message such as device not recognised appears. When a device is plugged into a USB port, the operating system searches for its driver and reports an error if none is found. USB drivers hold descriptors about each device: vendor ID (VID), product ID (PID) and a unique serial number, which the USB bus uses to ask a newly connected device what it is. A device with no serial number is treated as new every time it is plugged in, and two devices must never share a serial number.

Check yourself

8. 1. Give four examples of utility programs.
9. 2. Describe what heuristic checking does and why it is useful.
10. 3. Explain what happens to an infected file that is put into quarantine.
11. 4. Explain why files on a hard disk become fragmented over time, and how a defragmenter fixes this.
12. 5. State the three versions of a file recommended for total back-up security.

saklib.sk